R - EXERCISE OVERVIEW

Week 6 & 7

- ***27 – Databases and the role of SQL***

  No exercises.

- ***28 – Connecting to a database***

  Open a connection to the database (see the provided exercise file). Besides, use again the dataset `transactions.csv` (stored as `myData`).

  1. List all available tables that are stored in the database.
  2. Lists all variables of the table `demographics`.
  3. Store a copy of the table `myData` in the database and name this copy `myData_copy`.
  4. Remove `myData` from the database.
  5. Close the database connection.

- ***29 – Select operations in SQL***

  Open a connection to the database (see the provided exercise file).

  1. Use `dbGetQuery()` to execute separate SQL commands to show all the records in the `transaction` and the `demographics` table.
  2. Show the IDs of all customers who purchased on 06.06.2012. How many customers purchased on that date? Please Use `DISTINCT Customer` in your SQL query to account for duplicates.
  3. Calculate a new variable called `NetSales`, which should be the difference between purchase amount and cost per transaction. Check your results by looking at the whole table.

- ***30 – Aggregate operations in SQL***

  Open a connection to the database (see the provided exercise file).

  1. Show for each customer the latest transaction date (`LatestPurch`) as well as the maximum purchase amount per transaction (`MaxPurch`).
  2. Calculate the average purchase amount (`AvgPurch`) and average cost (`AvgCost`) per year across all customers.
  3. Write a query for a table that shows the number of transactions (`Transactions`), and the average purchase amount (`AvgPurch`) per customer per month. Please use `count(*)` to calculate the number of transactions.

- ***31 – Merge operations in SQL***

  Open a connection to the database (see the provided exercise file).

  1. Merge all transactions with the demographic details of the corresponding customers. What share of the overall purchase amount comes from `female` customers?
  2. Merge the transaction and the demographic data in a way, that the resulting table only shows the transactions on the day the customer joined the company.

- ***Excursus – Connecting to Apache Spark***

  1. Install Java 8 (http://java.com/en/download).
  2. Run the provided R script to do the following:
     a. Install the R packages `sparklyr` and install Spark.
     b. Start a Spark Session and import the CSV file `transactions.csv`.
     c. Use SQL in Spark to select all transactions with a cost lower than 30.